NORMAL UN-TOPOLOGY ON WEAKLY LINDELOF ATOMIC BANACH LATTICES

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Abstract. Kandić and Vavpetič ask for conditions ensuring that the unbounded norm topology of a Banach lattice is normal, and state that the problem is open even for atomic order-continuous Banach lattices. We prove that every atomic order-continuous Banach lattice which is Lindelöf in its weak topology has a Lindelöf, paracompact, normal un-topology. Thus the conclusion holds for weakly compactly generated lattices, including nonseparable $c_0(\Gamma)$. A product argument also covers a complementary sum of an atomic KB-band and a weakly Lindelöf atomic order-continuous band; this gives examples outside both classes previously handled in the source.

Status and source question

This is a candidate substantial partial answer to Problem 7.3 of Kandić–Vavpetič [1]. On PDF page 26 the source asks:

> Under which conditions on a Banach lattice X is the topological space (X, τ_{un}) normal?

It then says that the answer remains unknown even for order-continuous atomic Banach lattices. The source proves normality when τ_{un} is metrizable and when X is an atomic KB-space.

> the unbounded norm topology on X is a locally solid Hausdorff topology, so that (X, τ_{un}) is regular (see e.g. [Sch71]). Since (X, τ_{un}) is σ-compact, it is clearly a Lindelöf space, so that paracompactness of the unbounded norm topology finally follows from Morita's theorem [Dug66, Theorem VIII.6.5]. □
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> The following corollary is an easy consequence of [Dug66, Theorem VIII.2.2] and Theorem 7.1.
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> **Corollary 7.2.** *If X is an atomic KB-space, then (X, τ_{un}) is a normal topological space.*
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> The proof of Theorem 7.1 heavily depends on Morita's theorem from general topology. It would be of interest to find a functional analytical proof. We conclude the paper with the following open question.
>
> **Problem 7.3.** Under which conditions on a Banach lattice X the topological space (X, τ_{un}) is normal?
>
> The answer remains unknown even for order continuous atomic Banach lattices.

Definitions

For a Banach lattice X, the unbounded norm topology τ_{un} is the Hausdorff locally solid topology whose convergence is

$$x_\alpha \longrightarrow x \quad \Longleftrightarrow \quad \big\| \, |x_\alpha - x| \wedge u \, \big\| \longrightarrow 0 \quad (u \in X_+).$$

A space is *Lindelöf* when every open cover has a countable subcover. A Banach space is *weakly Lindelöf* when it is Lindelöf for $\sigma(X, X^*)$. A Banach space is weakly compactly generated (WCG)

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when a weakly compact subset has norm-dense linear span. Every WCG Banach space is weakly Lindelöf [3]; the wider WLD class is weakly Lindelöf as well [4].

Idea of the proof

Atomic order continuity turns the un-topology into the topology of pointwise convergence on the atomic coordinates. Those coordinates are bounded linear functionals, so the un-topology is coarser than the weak topology. This simple comparison transfers Lindelöfness from the weak topology to the un-topology; complete regularity then gives normality.

The mixed-band extension uses a second mechanism. An atomic KB-band is σ-compact in un-topology, while the complementary weakly Lindelöf band is Lindelöf in un-topology. Since compact times Lindelöf is Lindelöf, the product of these two factors is Lindelöf. This does not settle the general atomic order-continuous case: an arbitrary atomic Banach sequence ideal need be neither KB nor weakly Lindelöf and need not split into bands of these types.

The weakly Lindelöf criterion

Lemma 1. *If X is an atomic order-continuous Banach lattice, then*

$$\tau_{\mathrm{un}} \subseteq \sigma(X, X^*).$$

Equivalently, the identity map from $(X, \sigma(X, X^*))$ to (X, τ_{un}) is continuous.

Proof. Choose one normalized representative a from each atomic band and let f_a be its coordinate functional, so the corresponding band projection is $P_a = f_a \otimes a$. Kandić–Marabeh–Troitsky [2, Corollary 4.14] prove that, for every net (x_α),

$$x_\alpha \xrightarrow{\mathrm{un}} x \quad \Longleftrightarrow \quad f_a(x_\alpha) \longrightarrow f_a(x) \quad \text{for every atom } a.$$

Hence τ_{un} is precisely the initial topology generated by the family (f_a). Each f_a belongs to X^*, while the weak topology is generated by all of X^*. The asserted inclusion follows. □

Theorem 1. *Let X be an atomic order-continuous Banach lattice. If $(X, \sigma(X, X^*))$ is Lindelöf, then (X, τ_{un}) is Lindelöf, paracompact, and normal. In particular, this holds if X is WCG (or WLD).*

Proof. By Lemma 1, every τ_{un}-open set is weakly open. Thus every τ_{un}-open cover is a weakly open cover and has a countable subcover. So (X, τ_{un}) is Lindelöf.

The un-topology is a Hausdorff locally solid vector topology, hence completely regular; see, for example, the discussion before Theorem 7.1 in [1]. A regular Lindelöf space is paracompact, and every Hausdorff paracompact space is normal [5]. The final assertion follows from the classical weak-Lindelöf theorem for WCG spaces [3] (and the corresponding fact for WLD spaces [4]). □

Corollary 1. *For every set Γ, $c_0(\Gamma)$ has a normal un-topology. If Γ is uncountable, this supplies an atomic order-continuous example whose un-topology is nonmetrizable and which is not a KB-space.*

Proof. The lattice $c_0(\Gamma)$ is atomic and order continuous. It is WCG: if $(e_\gamma)_{\gamma \in \Gamma}$ is its unit-vector family, then

$$K = \{0\} \cup \{e_\gamma : \gamma \in \Gamma\}$$

is weakly compact. Indeed, K has its one-point-compactification topology, because every member of $c_0(\Gamma)^* = \ell_1(\Gamma)$ is non-negligible on only finitely many e_γ at any fixed threshold. The linear span of K is norm dense. Theorem 1 applies.

If Γ is uncountable, no countable family can generate a norm-dense ideal: the union of the supports of that family is countable. The metrizability criterion quoted in [1, Section 7] therefore shows that τ_{un} is nonmetrizable. Finally, for distinct γ_n, the increasing partial sums $\sum_{j \le n} e_{\gamma_j}$ are norm bounded but not norm convergent, so $c_0(\Gamma)$ is not a KB-space. □

A mixed-band upgrade

Theorem 2. *Let $X = Y \oplus Z$ be a decomposition into complementary projection bands. Assume that Y is an atomic KB-space and that Z is atomic, order continuous, and weakly Lindelöf. Then (X, τ_{un}) is Lindelöf, paracompact, and normal.*

Proof. For a finite band decomposition, un-convergence in X is equivalent to un-convergence of both band coordinates; this is the finite case of [2, Theorem 4.12]. Hence

$$(X, \tau_{\mathrm{un}}) \cong (Y, \tau_{\mathrm{un}}) \times (Z, \tau_{\mathrm{un}}).$$

By the atomic KB theorem [2, Theorem 7.5], every closed norm ball of Y is un-compact. Therefore $Y = \bigcup_{n \geq 1} nB_Y$ is σ-compact in τ_{un}. By Theorem 1, (Z, τ_{un}) is Lindelöf.

Write $Y = \bigcup_n K_n$ with each K_n compact. Each $K_n \times Z$ is Lindelöf because a compact space times a Lindelöf space is Lindelöf. The countable union $Y \times Z = \bigcup_n (K_n \times Z)$ is therefore Lindelöf. Complete regularity again yields paracompactness and normality. □

Corollary 2. *If Γ and Δ are uncountable, then*

$$X = \ell_1(\Gamma) \oplus c_0(\Delta)$$

(with any Banach-lattice norm realizing the displayed complementary bands) has a normal, nonmetrizable un-topology, although X is neither a KB-space nor weakly Lindelöf in its weak topology.

Proof. The first band is atomic KB, and the second is atomic, order continuous and WCG, so Theorem 2 applies. The $c_0(\Delta)$ band prevents X from being KB. Nonmetrizability follows as in Corollary 1, since a countable family has only countable total atomic support.

For completeness, $\ell_1(\Gamma)$ is not weakly Lindelöf when Γ is uncountable. Its unit vectors form an uncountable weakly closed discrete subspace: coordinate functionals isolate the points, and the constant-one functional together with finitely many coordinates separates any other vector from the family. A closed subspace of a regular Lindelöf space is Lindelöf, while a regular Lindelöf space cannot contain an uncountable closed discrete subspace. Thus the weak topology of X cannot be Lindelöf. □

Scope, verification, and novelty

The result supplies new sufficient conditions and explicit cases beyond the two classes in the source, but it is not a characterization. The full atomic order-continuous problem would follow from a normality theorem for arbitrary solid Banach sequence ideals in a Sigma-product, or from a decomposition into the two band types above. Neither bridge is established here; normality is not hereditary to arbitrary subspaces of the ambient Sigma-product.

The proof was audited implication by implication in `verification.md`. The accompanying script performs asset and marker checks only; no finite computation can verify this topological proof. A bounded local-index and web/arXiv/publisher search is recorded in `novelty.md`. It found no later solution of Problem 7.3 and no exact duplicate of these sufficient conditions, but the novelty claim remains provisional pending specialist database review.

References

[1] M. Kandić and A. Vavpetič, *On separability of the unbounded norm topology*, Positivity 27 (2023), Paper No. 43; arXiv:2105.03126.

[2] M. Kandić, M. A. A. Marabeh, and V. G. Troitsky, *Unbounded norm topology in Banach lattices*, J. Math. Anal. Appl. 451 (2017), 259–279; arXiv:1608.05489.

[3] J. Orihuela, *On weakly Lindelöf Banach spaces*, in Progress in Functional Analysis, North-Holland Mathematics Studies 170, North-Holland, 1992, pp. 279–291.

[4] S. A. Argyros and S. Mercourakis, *On weakly Lindelöf Banach spaces*, Rocky Mountain J. Math. 23 (1993), 395–446.

[5] J. Dugundji, *Topology*, Allyn and Bacon, Boston, 1966.